



3 Now, tear off two sheets of the paper towel and loosely scrunch them up. Push one crumpled paper towel into each end of the tube. The paper absorbs some of the high-pitched sounds coming from the tube, so music sounds less tinny.



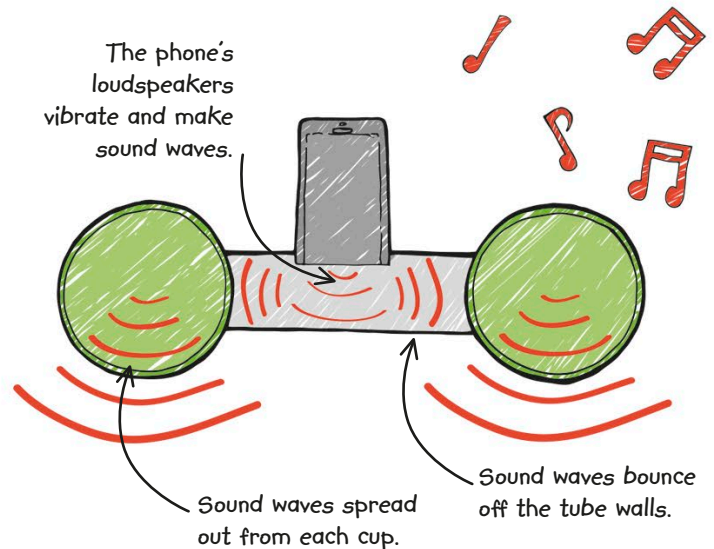
4 Push one end of the tube a little way into the hole you cut in one of the cups. You may need to use a small amount of force to secure it on. Then fit the other end of the tube into the second cup. You've nearly finished your speakers!

5 The only thing left to do is to paint your speakers in any colors you like. When the paint is dry, pop a phone into the slot, with its loudspeakers inside the tube. Then, sit back and enjoy the music!

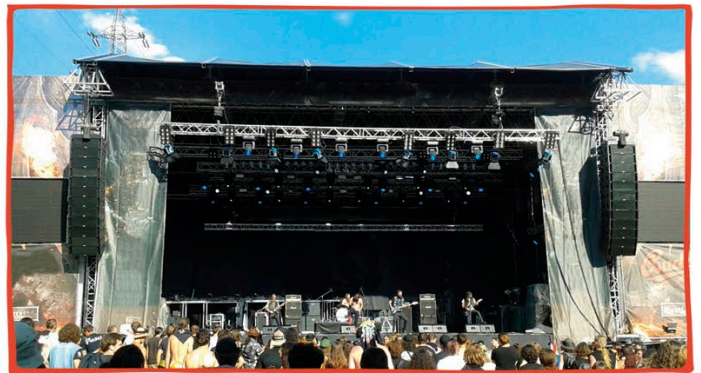


HOW IT WORKS

Sounds from the mobile phone come out of tiny loudspeakers that vibrate and disturb the air, sending sound waves spreading in all directions. When you put your phone inside your speakers, the sound waves bounce off the insides of the tube and the cups. So nearly all the sound is sent forward, toward your ears. The crumpled paper stops some of the higher-pitched sounds from getting through, but not the lower-pitched ones. This creates a clearer and warmer sound.



REAL WORLD SCIENCE CONCERT LOUDSPEAKERS



At a concert, powerful speakers sit on either side of the stage. Inside each speaker is a cone of paper that vibrates, driven by electrical signals from the equipment—such as an electric guitar—on stage. This produces sound waves that spread out in all directions. Some waves bounce off the back of the speakers and then forward to the audience.